Theory of Evolutionary Algorithms

1) History
2) (1+1) EA ≈ (1+1) ES
3) LINEAR PSEUDO-BOOLEAN FUNCTIONS

onemax
binary value

4) METHODS TO STUDY EA CONVERGENCE
   4.1) FITNESS-BASED PARTITIONS
   4.2) THE COUPON COLLECTOR PROBLEM
   4.3) EXPECTED MULTIPLICATIVE DISTANCE DECREASE
5) GEOMETRIC OPERATORS

History

"DARK AGES" ⟶ (ORIGINAL) SCHEMA THEORY

ex. individual: 1 0 1 1 0 1.

ex. schema: * * 1 1 * $\boxed{*}$ "DON'T CARE" symbol

A schema is a string in $\{0, 1, *\}^n$

Given a schema $\underline{s} \in \{0, 1, *\}^n$ and an individual $\underline{x} \in \{0,1\}^n$ we say that $\underline{x}$ contains $\underline{s}$ when:

$$\forall i \in \{1, \dots, n\}, \text{ if } s_i \in \{0,1\} \; s_i = x_i$$

$x = 0\,1\,0\,1$
$s = *\,1\,*\,1$ $\quad$ x contains s

$s' = 0\,0\,*\,*$ $\quad$ x does not contain s'

lenght of a schema as the difference between the index of the leftmost not "don't care" position and the index of the the rightmost not "don't care" position plus one

$\underbrace{0**1*1}_{\text{lenght of } 6}$ $\qquad$ $**\underbrace{010}_{\text{length of } 3}*$

what is the prob. of "breaking" a schema by performing one-point crossover?

the order of a schema is the number of not "don't care" symbols in it

$$\underbrace{0**1*1}_{\text{order } 3} \qquad \underbrace{**010*}_{\text{order } 3}$$

$$\left(1-\frac{1}{n}\right)^{\text{order}(s)}$$

"Informal results"

→ schemata with small order and small lenght where the average fitness of an individual that contains that schema is high survive more.

EA as Markov processes

will the EA converge to a stationary distributions where all populations containing a global optima have prob. 1?

s_i = population i

s_J = population J

$P(s_i \mid s_J)$ if we have mutation we have that this is positive

M is the transition matrix where

M_{iJ} is $P(s_i \mid s_J)$

M has only positive entries

only stationary distribution is the one satisfying $x = Mx$

BUT all populations will have positive probabilities

BUT if we add elitism...

... the process will converge to a distribution where only the populations with a global optimum have positive probabilities

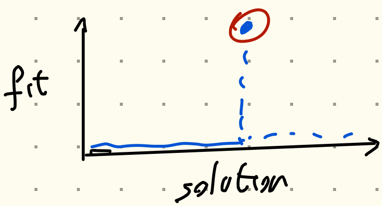
RUNTIME ANALYSIS

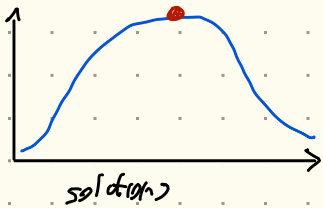
"INGREDIENTS"

1) AN EVOLUTIONARY ALGORITHM "A"

2) A PROBLEM OR A CLASS OF PROBLEMS "P"

[WHAT IS THE EXPECTED TIME FOR "A" TO FIND THE OPTIMUM FOR ANY PROBLEM IN THE CLASS "P"?]





(1+1) EA

```
x ← random {0,1}^n
forever {
    OFFSPRING ← MUTATE (x)
    x ← FITTEST (x, OFFSPRING)
}
```

bit flip with probability $1/n$

LINEAR PSEUDO-BOOLEAN FUNCTIONS

$$f: \{0,1\}^n \longrightarrow \mathbb{R}$$ "pseudo-Boolean"

$$x = (x_1, \dots, x_n) \in \{0,1\}^n$$

$$f(x) = w_1 x_1 + w_2 x_2 + \dots + w_n x_n = \sum_{i=1}^{n} w_i x_i$$

with $w_i \in \mathbb{N}$, , $w_i \neq 0$

LINEAR

GLOBAL OPTIMUM IS $\vec{1} = (1, 1, \dots, 1)$ when maximizing f

$\downarrow$

fitness $f(\vec{1}) = \sum_{i=1}^{n} w_i$

if $w_i = 1 \ \forall i \in \{1, \dots, n\}$ we have ONEMAX

$f(X) =$ # of 1s in X

if $w_i = 2^{i-1}$, i.e., $w_1 = 2^0$, $w_2 = 2^1$, $w_3 = 2^2$, ...

$f(x) =$ the binary number represented by x

Q1: what is the exp. opt. time of a (1+1) EA on ONEMAX?

Q2: what is the exp. opt. time of a (1+1) EA on any linear pseudo-Boolean function?

ONEMAX

fitness-based partitions

$S = \{0,1\}^n$

take a partition $A_1, \dots, A_m$ of S

i.e., $\bigcup_{i=1}^{m} A_i = S \quad \forall i,j \;\; i \neq j \;\; A_i \cap A_j = \emptyset$

$A_i <_f A_j \quad \forall a \in A_i \;\; \forall b \in A_j$

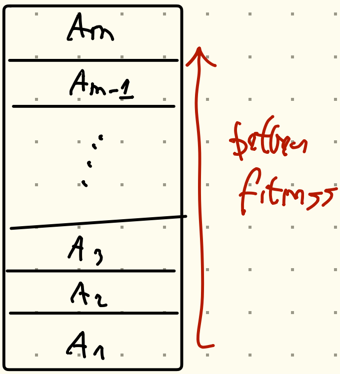
$f(a) < f(b)$

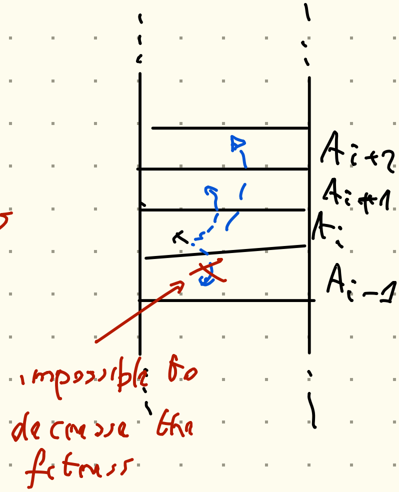
if:

1) $A_1 <_f A_2 <_f A_3 <_f \dots <_f A_m$

2) $A_m = \{ \vec{1} \}$ i.e., A_m contains only the global optimum

$A_1, \dots, A_m$ is a fitness-based partition





given $x \in \{0,1\}^n$

$p(x)$ = probability of "jumping" to a better partition.

i.e. if $x \in A_i$ $p(x)$ is the prob. of having an offspring in $A_{i+1} \cup A_{i+2} \cup \dots \cup A_m$

EXP. TIME TO "JUMP" IS $\dfrac{1}{p(x)}$

$p_i = \min_{a \in A_i} p(a)$ } remove the dependence on x

$\dfrac{1}{p_i}$ = upper bound on the exp. time to "jump" to better fitness level.

worst case when we jump:

→ we move from A_i to A_{i+1}

exp. time to reach the optimum is bounded above by:

$$\sum_{i=1}^{m-1} \frac{1}{p_i}$$

→ we need to find these probabilities

The onemax case

$w_i = 1 \quad f(x) = \sum_{i=1}^{n} x_i$

PARTITION

FITNESS

$$n+1 \text{ sets} \begin{cases} A_0 = \{0000\} & 0 \\ A_1 = \{0001, 0010, 0100, 1000\} & 1 \\ \vdots & \vdots \\ A_n = \{1111\} & n \end{cases}$$

$x \in A_{n-k}$ if k bits of x are zeros

Theorem The exp. opt. time for a (1+1) EA on the ONEMAX problem is

$$O(n \log n)$$

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prob. of moving from A_{n-k} to A_{n-k+1}

is $\leq p_{n-k}$

not flipping any other bit

$$\frac{k}{n}\left(1-\frac{1}{n}\right)^{n-1} \geq \frac{k}{e \cdot n}$$

prob. of flipping a 0 bit to a 1 bit

prob. of flipping exactly one 0 bit to a 1 bit

$$p_{n-k} \geq \frac{k}{e \cdot n} \rightsquigarrow \frac{1}{p_{n-k}} \leq \frac{e \cdot n}{k}$$

$$\sum_{k=1}^{n} \frac{1}{p_{n-k}} = \sum_{k=1}^{n} \frac{e \cdot n}{k} = e \cdot n \cdot \sum_{k=1}^{n} \frac{1}{k} =$$

$$= O(n \cdot \log n)$$

∎

(1+1) EA on ONEMAX requires $O(n \log n)$ time (on expectation) to reach the optimum

Coupon collector problem

Th. Expected opt. time for (1+1) EA on every linear pseudo Boolean function is $\Omega(n \log n)$.

0	1	0	0	1	1	0	1	1

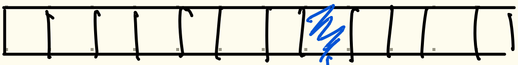
1 1 1 1 1 1 0 1 1

Chernoff Bound

$\frac{n}{2}$ bits set to 1 (on expectation)

$\leq \frac{2}{3} n$ bits set to 1 has prob. $1 - e^{-\Omega(n)}$

$\Rightarrow$ with prob. $1 - e^{-\Omega(n)}$ at least $\frac{n}{3}$ of the bits are set to 0.



prob. that after t time steps the position has been flipped

$$1-\left(1-\frac{1}{n}\right)^{t}$$

this bit has been flipped at least once.

the bit has not been flipped after t time steps

$$1-\left(1-\left(1-\frac{1}{n}\right)^{t}\right)^{n/3}$$

we still have bits to flip

probability that we flipped the these bits at least once

MAGIC HERB → $t=(n-1)\log n$

we select

$$1-\left(1-\left(1-\frac{1}{n}\right)^{(n-1)\log n}\right)^{n/3} \geq 1-e^{-1/3}$$

$$\left(1-e^{-\Omega(n)}\right)\left(1-e^{-1/3}\right) \geq 1-e^{-\Omega(n)}-e^{-1/3} = \Omega(1)$$

we have to flip at least $n/3$ bits

we still have bits to flip after $(n-1)\log n$ time steps

it is a positive constant

Hence, we need at least $(n-1)\log(n)$, i.e., $\Omega(n \log n)$ steps to flip all zero bits.

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onemax

$O(n \log n)$ $\Omega(n \log n)$

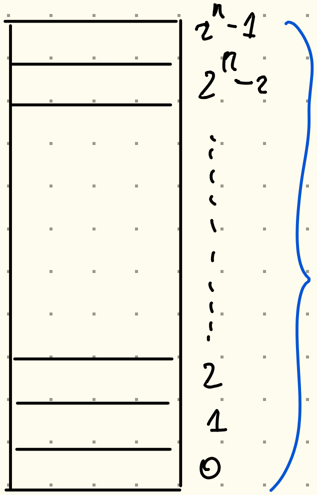
Holds for ONEMAX

Holds for all union pseudo-Boolean functions.

$\Theta(n \log n)$

$$f(x) = \sum_{i=1}^{n} 2^{n-i} x_i \qquad f(\overset{8}{0}\overset{4}{1}\overset{2}{0}\overset{1}{1}) = 5$$

fitness levels



2^n fitness levels

p_i = prob. of jumping to a higher fitness level

$$\sum_{i=0}^{2^n-2} \frac{1}{p_i} \rightsquigarrow \text{even if } p_i = 1$$

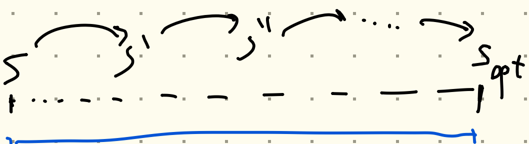
we have an exp. bound.

Exp Multiplicative Distance Decrease

to prove that the upper bound is actually

$$O(n(\log n + \log w_{max}))$$

maximum weight of the function.



distance in fitness

$f(s_{opt}) - f(s)$

$O = \{o_1, o_2, \dots, o_r\}$ set of operations

where:

1) each operation has the same probability of being applied.

2) there exists a subset of operations that, when applied, produces the optimum.

$O = \{$ flip bit 1, flip bit 2, $\vdots$ flip bit n $\}$

$$\frac{1}{n}\left(1 - \frac{1}{n}\right)^{n-1}$$

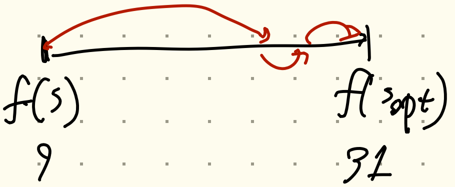
prob for each element (operation) in O

$\forall s \in \{0,1\}^n$ there exists $O' \subseteq O$ such that by performing all operations in O' we reach $\vec{1}$, i.e., the optimum

$$\begin{array}{cccccc} & 16 & 8 & 4 & 2 & 1 \\ s = & 0 & 1 & 0 & 0 & 1 \\ & 1 & 2 & 3 & 4 & 5 \end{array}$$

$O' = \{$ flip bit 1, flip bit 3, flip bit 4 $\}$

$f(s) = 9 \quad f(s_{opt}) = f(\vec{1}) = 31$



$$f(s_{opt}) - f(s) = 31 - 9 = \\ = 22$$

$$\underbrace{\frac{f(s_{opt}) - f(s)}{n}}$$

expected decrease in distance when we perform an operation in O.

n: "worst case": we need all operations

$$d = f(s_{opt}) - f(s)$$

Exp. distance from s_{opt} of the offspring of s after applying an operation in O.

$$d - \frac{d}{n} = \left(1 - \frac{1}{n}\right) d$$

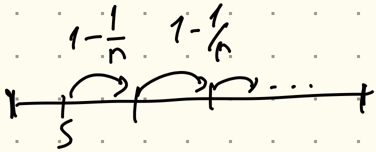
to remove the dependency from s we start from the point at maximum distance from s_{opt}

$$d_{max} = \max_{s \in \{0,1\}} f(s_{opt}) - f(s)$$

$$\left(1 - \frac{1}{n}\right) d \leq \left(1 - \frac{1}{n}\right) d_{max}$$

AFTER t time steps

$$\left(1-\frac{1}{n}\right)^t d_{max}$$



At a certain point you get

$$\left(1-\frac{1}{n}\right)^t d_{max} \leq \frac{1}{2}$$

- the distance can only be a natural number
- Hence the distance is 0 at least half the times

optimum reached

TODO: find a value of t for which

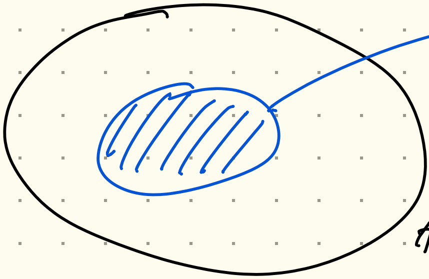
$$\left(1-\frac{1}{n}\right)^t d_{MAX} \leq \frac{1}{2}$$

$$\Rightarrow \quad t = c \cdot n \cdot \log d_{MAX}$$

constant

Exp. number of operations of O to reach the optimum is $O(n \cdot \log d_{max})$

in the case of linear pseudo-Boolean functions $r = n$, so $O(n \log d_{max})$



mutations in O

All possible operation

αr ← number of operations in O

prob. of performing an operation in O

$\frac{1}{\alpha \cdot r}$ → exp. time to apply an operation in O

$$O\left(\underbrace{\frac{1}{\alpha \cdot \cancel{r}}}_{\text{time between operations in } O} \cdot \overbrace{\cancel{r} \cdot \log d_{max}}^{\text{number of operations in } O \text{ that we need to perform}}\right)$$

$$O\left(\alpha^{-1} \log d_{max}\right)$$

$$\alpha = \frac{1}{n}\left(1-\frac{1}{n}\right)^{n-1} \geq \frac{1}{e \cdot n}$$

$$O(\alpha^{-1} \log d_{max}) = O(e \cdot n \cdot \log d_{max}) =$$

$$= O(n \log d_{max})$$

$$d_{max} = \sum_{i=1}^{n} w_i \leq \sum_{i=1}^{n} w_{max} = n \cdot w_{max}$$

$$O(n \cdot \log n \cdot w_{max}) =$$

$$O(n(\log n + \log w_{max}))$$

exp. opt. time to reach the optimum for any linear pseudo-Boolean function.

ONEMAX $\quad w_{MAX} = 1 \qquad O(n \log n)$

Binary value $\quad w_{MAX} = 2^{n-1} \quad O(n(\log n + \log 2^{n-1})) =$

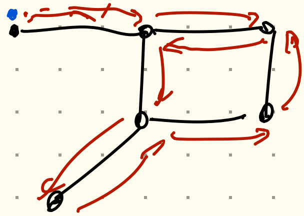
$$= O(n \log n + n(n-1)) = O(n^2)$$

* Covering time of Graphs

- example of non linear - pseudo Boolean fitness function

- No free lunch theorem

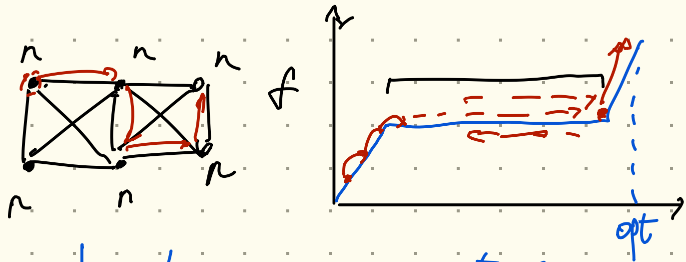
$G = (V, E)$



What is the exp. time to cover the graph

visit every node

$$f(x) = \begin{cases} n & \text{if the number of ones is more than } n/2 \\ \dots & \end{cases}$$

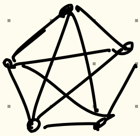


Thm upper bound on covering time

Given an undirected graph $G = (V, E)$ with $|V| = n$ vertices and $|E| = m$ edges the cover time is upper bounded by $O(2m(n-1))$

n vertices

cover $O(n^2)$

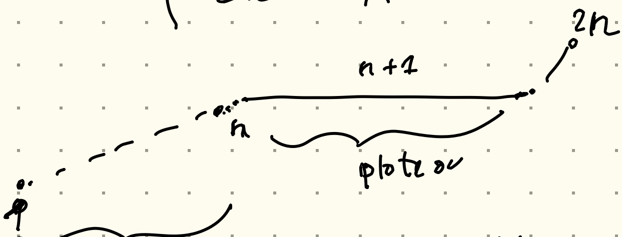


n vertices

$O(n^2)$ edges cover $O(n^3)$

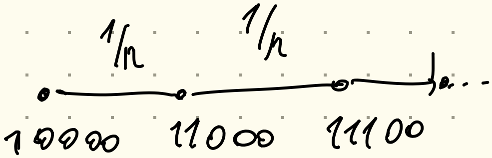
SPC = shortest path with constant values

$$SPC(x) = \begin{cases} |x|_0 & \text{number of zeros} \quad x \notin \{1^k 0^{n-k} : 0 \le k \le n\} \\ n+1 & x \in \{1^k 0^{n-k} : 0 \le k < n\} \\ 2n & x = 1^n \end{cases}$$



when we are not of the form $1^k 0^{n-k}$

$O(n \log n)$



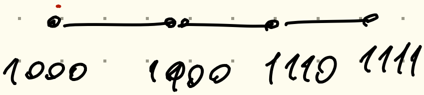
$\cancel{10100}$

$\uparrow$

$10000 \longrightarrow 11000$

$\downarrow$

$\cancel{10010}$



$$O(2|E|(|V|-1)) =$$
$$= O(2(n-1)(n-1)) =$$
$$= O(n^2)$$

$$\left(\begin{array}{c}\text{prob. of taking} \\ \text{an edge in the} \\ \text{graph} = \frac{1}{n}\end{array}\right)^{-1} \cdot \begin{array}{l}\text{number of steps in} \\ \text{the graph}\end{array}$$

$$\left(\frac{1}{n}\right)^{-1} \cdot O(n^2) = n\,O(n^2) = O(n^3)$$

$$O(\underbrace{n\log n}_{\text{reaching the plateau}} + \underbrace{n^3}_{\text{plateau}}) = O(n^3)$$

$$NEEDLE = \begin{cases} 0 & \text{if } x \neq 1^n \\ 1 & \text{if } x = 1^n \end{cases}$$

$\underbrace{\text{No Free Lunch}}_{\text{NFL}}$ Theorem

Let S and R be finite sets and $F = R^S$ the set of all function

$$\underbrace{f}_{\text{fitness function}} : \underbrace{S}_{\text{solutions}} \longrightarrow \underbrace{R}_{\text{fitness values}}$$

Let A and A' be two stochastic opt. algorithms visiting each solution at most once.

Then the average number of fitness evaluations across all functions in F is the same for A and A'